

**Understanding the Association Between Childhood Puzzle/Board Game Play with Memory
Recall Performance on Shallow and Deep Processing Condition**

Prutha Patel

Department of Psychology, University of California, Davis

PSC 146 - The Development of Memory

Dr. Simona Ghetti

March 8, 2026

Introduction

Board games and puzzles play an important role during childhood in dictating executive functioning of working memory, integration of information and planning action. Playing board games and solving puzzles during childhood provides cognitive, behavioral and social training. Incorporating play of board games can help predict academic performance in reading, mathematics, and science achievement during elementary schooling (Thorell et al, 2009; Willoughby et al., 2012; Morgan et al, 2019; Traverso et al., 2019). Small intervention of play plan and circle games among kindergarten children can help improve attention and develop self-regulation (Shiu et al., 2018). Understanding the long-term positive benefits of board games and solving puzzles can help incorporate these cognitive learning strategies especially in school settings.

Intervention of board games can help improve academic skills and executive functioning. In the study conducted by Vita-Barrull et al. (2023), performed a randomized control study experiment among 99 students in a rural school. The study had an experiment group, the Conectar Jugando Group that received intervention of playing with commercialized board games for a duration of 6 weeks. The control group had no intervention and attended the regular classes during the study. Both the test group had a pre-assessment and post-assessment that measured updating working memory of executive functions and administered PREDISCAL test to assess math skills and academic reading. It was observed that the Conectar Jugando Group increased the percentage of correct answers in the working memory test compared to the control group that had lessened the percentage of correct answers on the task. Both groups performed well on the academic tasks, however the Conectar Jugando Group had a significantly greater score in calculus than the control group.

A similar study involved modern board game playing intervention among 522, first to sixth grade students and assessed their executive function and academic skill. The experimental group of second to sixth grade play groups had significant improvement on the executive function of updating tasks compared with the control group (Vita-Barrull et al, 2024). The study suggested incorporation of board games during regular school days to improve cognitive training and academic performance.

Veraksa et al. (2022) studied the use of different play games in assessing improvement in executive functions (EFs) of working memory, shifting and inhibition. The study involved 199 participants between the age of 52–69 months with preliminary assessment, 7 weeks of interventions and testing after the intervention. The experimental group was divided into five interventions: games with rules (board games), digital game, role play (free play, adult-directed play and child-directed play). In the group of games with rule a significant improvement was observed in visuospatial working memory than the control group. The intervention of game with rules had the highest response in visuospatial working memory development compared with other experimental interventions.

In another research experiment conducted by Estrada-Plana et al. (2024) tested the effectiveness of game-based intervention among 234 participants of third and fourth grade children on improving memory and math skills. The participants were divided into three groups: playing memory board games, playing mathematical board games and a control group. Both the game intervention: playing memory board games and playing mathematical board games had significantly improved short-term memory and working memory compared to the control group that had no intervention. These two groups had also displayed increased mathematical skills of

number operations and number ranking. Playing board games had a positive influence on cognitive skills and mathematical skills.

Previous research studies have focused on play intervention and observing its effect during elementary and middle school. In this study, I focus my research on the association between puzzle/board games played in childhood and working memory during adulthood. Is childhood puzzle and board game play associated with better word recall performance in adulthood under shallow processing and deep processing conditions? I hypothesize that adults who played puzzles/board games during childhood would recall a higher number of words on shallow and deep processing memory tasks than adults who as children didn't play puzzles/board games during childhood. I conducted working memory tests under shallow processing and deep processing conditions from young adults through questionnaires.

Methods

Participants: The study consisted of 220 individuals between 18 and 55 years of age (age: $M = 22.4$, $SD = 5.69$). Participants' self-reported gender included female ($N = 130$; age: $M = 22.5$, $SD = 5.97$), male ($N = 84$; age: $M = 22.4$, $SD = 5.44$), and non-binary ($N = 4$; age: $M = 21.5$, $SD = 2.38$). Participants' race/ethnicity was distributed as White ($N = 52$), Asian ($N = 92$), African American ($N = 3$), Hispanic ($N = 45$) and other ($N = 25$). Participants' primary language reported was English ($N = 168$), Spanish ($N = 16$), Chinese ($N = 12$), and other ($N = 23$). Participants' highest education level of all parents/guardians was distributed as some high school ($N = 7$), high school graduate or GED ($N = 27$), associate degree or two years of college or less ($N = 33$), bachelor's degree ($N = 77$), professional or doctoral degree ($N = 64$) and other ($N = 10$). Out of the 220 participants, 186 of them spoke a language other than English. Two of the participants were excluded from the analysis due to missing data.

Material: The convenience sample was identified by students of PSC 146, The Development of Memory class at UC Davis. To collect the data, each student interviewer administered the same questionnaire to the participant which consisted of six sections. The first section of the questionnaire asked questions regarding the participant's date of birth, age, gender, race/ethnicity, primary language, whether or not they speak a second language, age of exposure of second language, parents/guardians education levels, household yearly income, number of siblings and participant's birth order. The second section included word tasks of deep processing memory and shallow processing memory which asked the participants to remember as many words as they could after the student interviewer read the list of 15 words. The third section asked about the age of earliest memory remembrance, describing the earliest memory and earliest childhood memory statement that was answered by a 5-point scale of "very clear", "somewhat clear", "neutral", "somewhat fuzzy" and "very fuzzy". The fourth section included statements about childhood interest and activities that were answered by a 6-point scale "every day", "several times a week", "once or twice a week", "less than once a week", "once or twice a month", "never or almost never". Fifth and sixth sections of the questionnaire consisted of statements about participants' childhood learning strategies at school and participants' childhood conversation with parents which was answered using the above mentioned 5-point scale.

Procedure: Student researchers conducted the interview either in-person, over the phone or over Zoom. They also recorded whether the location of the interview was at home, outdoors or other. Researchers were required to provide debriefing of the study at the end of the questionnaire to the participant explaining the goals and means of this research study.

Statistical Analysis: The data was analyzed using Jamovi software program to create a spearman's rho and correlation coefficient to determine the relationship between the variables to

test the hypothesis that spending more time playing puzzles/board games in childhood improved memory retention. I used participants' responses to the question of how often they played puzzle/board game in childhood as the independent variable and each participants' correct number of recalled words on the deep processing and shallow processing memory test as the dependent variable. The frequency of playing puzzle/board games was measured on a 6-point scale. On the 6-point scale, the participants' response of every day will be codified as "6", several times a week as "5", once or twice a week as "4", less than once a week as "3", once or twice a month given a "2", and never or almost never are given a "1". I conducted a correlation to compare puzzle/board game activity levels and memory retention scores.

Results

A Spearman's rho correlation coefficient was computed to assess the linear relationship between puzzle/board game and memory retention on deep and shallow processing tasks. There was a positive correlation between the childhood characteristics of playing puzzle/board games and deep processing task, $r(df) = 0.144$, $p < 0.05$. In figure 1, the x-axis describes one's frequency of playing puzzle/board game in childhood and the y-axis describes each participants' correct recalled word on the deep processing task.

Similarly, a positive correlation was also observed between playing puzzle/board games and shallow processing task, $r(df) = 0.186$, $p < 0.01$. In figure 2, the x-axis describes one's frequency of playing puzzle/board game in childhood and the y-axis describes each participants' correct recalled word on the shallow processing task. Both the correlation demonstrated that this is a statistical significance. The observed correlation supports my hypothesis that frequent involvement of playing with puzzle/board games in childhood improved better word recall performance in adulthood under shallow and deep processing conditions.

Discussion

The goal of this study was to examine the association between early childhood time spent playing puzzle/board games and working memory during adulthood.

When looking at the performance of participants on word recall memory under shallow processing and deep processing strategies, it was observed that participants with frequent involvement of puzzle/board games during childhood performed better than those who had less frequent playing with puzzle/board games. The correlation between the variables of time spent playing puzzle/board games and word recall memory test had statistically significant differences under both shallow processing and deep processing strategies. The results from this study supported previous research that looked at the benefits of playing board games in early childhood. Here my research study expanded on the possibility of a positive long-term effect of playing with board games/puzzles on the working memory in adulthood. The study conducted by Ahmed et al. (2019), examined the executive function of working memory, sustained attention, delay to gratification and planning among 1273 participants that were tested at 54 months and 15 years of age. The longitudinal study had observed, among different components of executive function tested in the study, only working memory at the age of 54 months had significantly predicted working memory at 15 years of age in these participants. This study supports my idea that working memory early in childhood has long-term positive effects later during adolescence and possibility into adulthood.

The research question on understanding the relationship of puzzle/board games playing and working memory in adulthood still needs more research to be examined. One possible future study can be adapted from Ahmed et al. (2019), which can continue the longitudinal study of measuring working memory at adulthood. The study can further manipulate the types of games

that can be administered among the participants. The variable that will be measured in this follow-up study would be different game play and the measurement of working memory at the age of 54 months, 15 years and 25 years of age.

The limitation of the present study is that the surveys acquired by the student research are at convenience and not randomized sample selection. The participants might have been biased to perform well on these questions as they know the experimenter, possibly inputting some biases of the participants performing better. The sample collected by the student research is not the best representation of different race/ethnicity. In the further study, involving randomized sample selection and single-blind study, where the participants don't know the experimenter can be incorporated to remove biases in the study.

Understanding the role of puzzle/board games can essentially help shape the working memory of humans for better performance. Puzzle/board games help improve academic achievement and develop self-regulation which are key factors in a student. Further research into this topic can help benefit the academic setting and parental interaction with children for improving working memory.

Figures

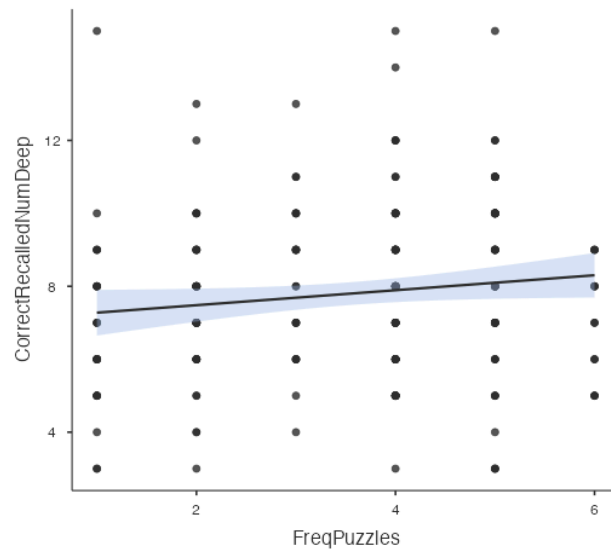


Figure 1: Correlation between puzzle/board game frequency and correct recalled words on the deep processing task

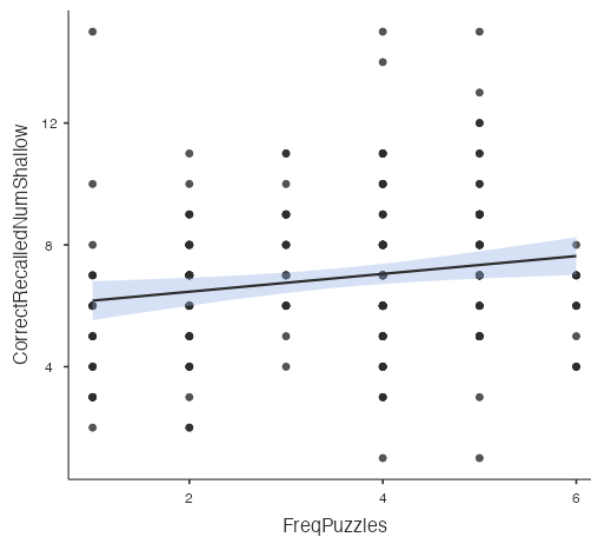


Figure 2: Correlation between puzzle/board game frequency and correct recalled words on the shallow processing task

References

- Traverso, L., Viterbori, P., & Usai, M. C. (2019). Effectiveness of an Executive Function Training in Italian Preschool Educational Services and Far Transfer Effects to Pre-academic Skills. *Frontiers in psychology*, *10*, 2053. <https://doi.org/10.3389/fpsyg.2019.02053>
- Ahmed, S. F., Tang, S., Waters, N. E., & Davis-Kean, P. (2019). Executive function and academic achievement: Longitudinal relations from early childhood to adolescence. *Journal of Educational Psychology*, *111*(3), 446–458. <https://doi.org/10.1037/edu0000296>
- Morgan, P. L., Farkas, G., Hillemeier, M. M., Pun, W. H., & Maczuga, S. (2019). Kindergarten Children's Executive Functions Predict Their Second-Grade Academic Achievement and Behavior. *Child development*, *90*(5), 1802–1816. <https://doi.org/10.1111/cdev.13095>
- Di Lieto, M. C., Pecini, C., Castro, E., Inguaggiato, E., Cecchi, F., Dario, P., Cioni, G., & Sgandurra, G. (2020). Empowering Executive Functions in 5- and 6-Year-Old Typically Developing Children Through Educational Robotics: An RCT Study. *Frontiers in psychology*, *10*, 3084. <https://doi.org/10.3389/fpsyg.2019.03084>
- Willoughby, M. T., Kupersmidt, J. B., & Voegler-Lee, M. E. (2012). Is preschool executive function causally related to academic achievement?. *Child neuropsychology : a journal on normal and abnormal development in childhood and adolescence*, *18*(1), 79–91. <https://doi.org/10.1080/09297049.2011.578572>
- Shiu, S. P., Wang, S. H., & Chen, Y. J. (2020). The effects of a small scale intervention to develop self-regulation in kindergarten children. *Early Child Development and Care*, *190*(3), 333–347. <https://doi.org/10.1080/03004430.2018.1471474>

- Veraksa, A., Sukhikh, V., Veresov, N., & Almazova, O. (2022). Which play is better? Different play types and development of executive functions in early childhood. *International Journal of Early Years Education*, 30(3), 560–576.
<https://doi.org/10.1080/09669760.2022.2091979>
- Moya-Higueras, J., Solé-Puiggené, M., Vita-Barrull, N., Estrada-Plana, V., Guzmán, N., Arias, S., Garcia, X., Ayesa-Arriola, R., & March-Llanes, J. (2023). Just Play Cognitive Modern Board and Card Games, It's Going to Be Good for Your Executive Functions: A Randomized Controlled Trial with Children at Risk of Social Exclusion. *Children (Basel, Switzerland)*, 10(9), 1492. <https://doi.org/10.3390/children10091492>
- Estrada-Plana, V., Martínez-Escribano, A., Ros-Morente, A., Mayoral, M., Castro-Quintas, A., Vita-Barrull, N., Terés-Lleida, N., March-Llanes, J., Badia-Bafalluy, A., & Moya-Higueras, J. (2024). Benefits of Playing at School: Filler Board Games Improve Visuospatial Memory and Mathematical Skills. *Brain sciences*, 14(7), 642.
<https://doi.org/10.3390/brainsci14070642>
- Vita-Barrull, N., Estrada-Plana, V., March-Llanes, J., Guzmán, N., Fernández-Muñoz, C., Ayesa, R., & Moya-Higueras, J. (2023). Board game-based intervention to improve executive functions and academic skills in rural schools: A randomized controlled trial. *Trends in neuroscience and education*, 33, 100216. <https://doi.org/10.1016/j.tine.2023.100216>
- Vita-Barrull, N., Estrada-Plana, V., March-Llanes, J., Sotoca-Orgaz, P., Guzmán, N., Ayesa, R., & Moya-Higueras, J. (2024). Do you play in class? Board games to promote cognitive and educational development in primary school: A cluster randomized controlled trial. *Learning and Instruction*, 93, Article 101946.
<https://doi.org/10.1016/j.learninstruc.2024.101946>

Cravet, E., & Usai, M. C. (2024). A Short Game-Based Intervention for Enhancing Executive Functions at School: a Pilot Study. *Journal of Cognitive Enhancement*, 8(3), 185–190.

<https://doi.org/10.1007/s41465-024-00301-5>

Thorell, L. B., Lindqvist, S., Bergman Nutley, S., Bohlin, G., & Klingberg, T. (2009). Training and transfer effects of executive functions in preschool children. *Developmental Science*, 12(1), 106–113. <https://doi.org/10.1111/j.1467-7687.2008.00745.x>